

AI PROTOTYPE · FINAL MASTER PROJECT

Exposed Rebar Inspection with YOLOv8 Segmentation

AI-assisted detection and pixel-level delineation of exposed reinforcement in concrete-inspection images

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SLIDE 1

Problem and objective

Engineering problem

Loss of concrete cover exposes reinforcement and raises the risk of corrosion and durability loss.

Current practice

Visual inspection of large photo sets is slow and depends on individual judgement.

Objective

Detect and segment visible exposed rebar in inspection photographs.

Intended role

Inspection decision support — not autonomous structural diagnosis.

SLIDE 2

The AI task: instance segmentation

Input

A photograph of a reinforced-concrete element taken during inspection.

Output

A bounding box plus a pixel-level segmentation mask for each exposed-rebar region.

Why masks

Masks describe the extent and geometry of the damaged area, not only its location.

Target class: `exposed_rebar` — Positive: reinforcing steel visible because cover is absent or broken.
Excluded: cracks, rust staining without visible steel, spalling without visible bars, shadows, mesh, pipes.

SLIDE 3

Method and workflow

Model: YOLOv8n-seg instance-segmentation network, trained on an annotated exposed-rebar image dataset (Roboflow rebar-exposure-qm020, v4; 1,547 training / 168 validation images).

- 1 Dataset preparation and annotation check
- 2 Model training (30 epochs)
- 3 Validation on unseen images
- 4 Inference on new inspection photos
- 5 Evidence, weights and code archived in GitHub

SLIDE 4

Training configuration

Parameter	Value
Task	Instance segmentation
Model	YOLOv8n-seg (yolov8n-seg.pt)
Epochs	30
Image size	640 × 640 px
Batch size	8
Hardware	NVIDIA Tesla T4 GPU
Random seed	42
Training time	≈ 20 minutes

Reproducibility

- Single Colab notebook reproduces training, validation and inference.
- Fixed seed 42 and recorded hyper-parameters.
- Selected checkpoint best.pt stored in the repository.
- Curves and epoch-level metrics exported as results.png and results.csv.

SLIDE 5

Validation results after 30 epochs

Metric	Bounding boxes	Segmentation masks
Precision	0.805	0.808
Recall	0.703	0.717
mAP@50	0.772	0.762
mAP@50–95	0.529	0.470

Reading the numbers

Mask metrics are the primary result: the task requires spatial delineation, not only localisation. The lower mAP@50–95 shows masks are coarse at strict overlap thresholds.

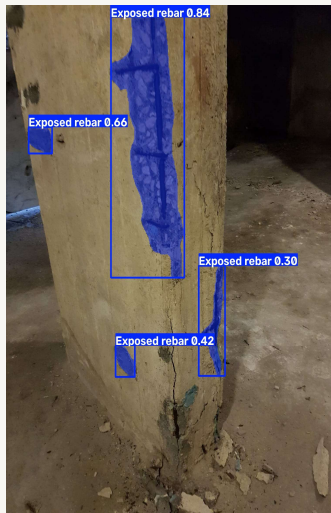
Validation set: 168 images, 229 annotated instances.

Training and validation curves (results/curves/results.png) show losses decreasing and validation metrics improving steadily through the 30 epochs — insert the curve image here in the final deck.

SLIDE 6

Prediction evidence on new inspection images

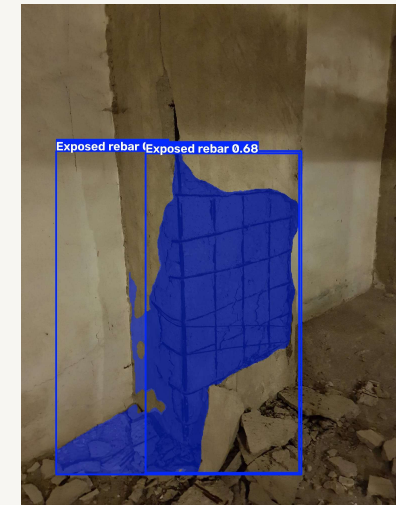
The trained checkpoint was applied to reinforced-concrete photographs outside the training workflow. Blue rectangles mark detected regions; blue masks mark the predicted pixel-level extent of the exposed reinforcement.



Local exposed reinforcement on a column or wall



Exposed bars on an overhead concrete element



Damaged wall or column area

Image files stored in results/evidence/ in the project repository.

Engineering value, limitations and next steps

Value

- Faster first-pass review of site photographs
- Consistent visual flagging of suspected exposure
- Supports damage inventories and inspection records

Limitations

- Sensitive to lighting, shadows, occlusion and viewpoint
- Can confuse cracks, staining and spalling with rebar
- Says nothing about corrosion severity or repair need

Next steps

- Validate on a larger independent set of site images
- Cover more element types and site conditions
- Keep qualified-engineer review before any decision

A detection is an inspection flag, not a structural diagnosis. Outputs must be reviewed by a qualified structural engineer.